Polarization of Light Waves

1. Definition

Polarization is the phenomenon in which the vibrations of light waves are restricted to a particular direction perpendicular to the direction of propagation of light.

It proves that light is a transverse wave.

2. Principle

Light waves are transverse electromagnetic waves. In unpolarized light, the electric field vibrates in different directions perpendicular to the direction of propagation. When light passes through a polarizer, its vibrations are restricted to a particular direction, producing plane-polarized light.

3. Types of Light

1. Unpolarized Light:
Light in which the electric field vibrates in many different directions perpendicular to the direction of propagation.

2. Plane-Polarized Light:
Light in which the electric field vibrates in only one fixed direction perpendicular to the direction of propagation.

4. Methods of Polarization

1. Polarization by Reflection: Light becomes partially or completely polarized when reflected from a surface at a particular angle.
2. Polarization by Refraction: Light can become polarized when it passes through refracting materials under suitable conditions.
3. Polarization by Double Refraction: Certain crystals split incident light into two rays with different polarization directions.
4. Polarization by Polaroid: A Polaroid sheet transmits light vibrating along its transmission axis and absorbs or blocks the perpendicular component.

5. Brewster's Law

When unpolarized light is incident at Brewster's angle, the reflected light is completely plane-polarized.

Formula:

$\mu = \tan i_p$

Where:

- μ = refractive index of the reflecting medium relative to the incident medium
- i_p = Brewster's angle

At Brewster's angle, the reflected and refracted rays are perpendicular to each other.

$i_p + r = 90°$

Where r is the angle of refraction.

6. Malus's Law

Malus's law gives the intensity of plane-polarized light transmitted through an analyzer.

Formula:

$I = I_o \cos^2 \theta$

Where:

- I = transmitted intensity
- I_o = intensity of polarized light incident on the analyzer
- θ = angle between the transmission axes of the polarizer and analyzer

Special cases:

- When $\theta = 0°$, $I = I_o$ (maximum intensity).
- When $\theta = 90°$, $I = 0$ (minimum intensity for ideal polarizers).

7. Applications of Polarization

1. Used in Polaroid sunglasses to reduce glare.
2. Used in photography to reduce unwanted reflections.
3. Used in LCD screens.
4. Used in stress analysis of transparent materials.
5. Used in 3D movie glasses.
6. Used to study the optical properties of materials.

8. Conclusion

Polarization is an important property of transverse waves. It demonstrates the transverse nature of light and has several applications in optical instruments and everyday devices.

Important formulas:

- Brewster's law: $\mu = \tan i_p$
- Malus's law: $I = I_o \cos^2 \theta$
- Brewster's angle condition: $i_p + r = 90°$